24 An ambulance siren emits a sound with a single frequency *f*.

The ambulance travels towards, passes close to, and then travels away from a stationary observer.

Which statement describes the frequency of the sound detected by the observer as the ambulance **passes** the observer?

A equal to *f* and decreasing

B equal to *f* and increasing

C greater than *f* and constant

D less than *f* and constant